MATH 540. HW # 9.

Prob. 1. (2RT, ch 7. 1.1)

Show that all linear f-ns:

(1) $$u = a_1 x_1 + a_2 x_2 + \dots + a_n x_n + a_0$$

are <u>harmonic</u> in $\mathbb{R}^n$.

?1 $C^2(\Omega)$

Show 2 things: (1) that they are sufficiently smooth (in C^0) & (2) that they satisfy the Laplace eq-n. (Recall the property 1 of harmonic f-ns: if $u(x,y)$ is cont-s & satisfies the Laplace eq-n, then $u(x_1, \dots x_n)$ is $C^\infty(\Omega)$. So in particular $u \in C^2(\Omega)$, & thus it is sufficient to show that $u(\underline{x})$ is con-s ($C^0(\Omega)$?2) & satisfies the Laplace eq-n.)

?3 Prop 2.5?

Sol-n: By the hint above, let me try to show (1) smoothness & (2) $\Delta u = 0$:

(1) $u(\underline{x}) = a_1 x_1 + \dots + a_n (x_n) + a_0$ is a poly-L, so it's, indeed, from $C^\infty(\mathbb{R}^n)$. i.e.

(2) $u \in C^\infty(\mathbb{R}^n)$. ✓

(2) $\dfrac{\partial u}{\partial x_i} = a_i = \text{const.}$ for $i = 1, 2, \dots n$ & thus

$\dfrac{\partial^2 u}{\partial x_i^2} = \dfrac{\partial}{\partial x_i}[a_i] = 0 \quad \forall\, i = 1, 2, \dots n,$ then

(3) $$\Delta u = \sum_{i=1}^{n} \frac{\partial^2 u}{\partial x_i^2} = 0 + 0 + \dots + 0 = 0 \equiv 0 \checkmark$$

Hence, the given u in (1) is harmonic. ∎